

HiMCM 2020 Problem A — Summary Sheet

Team XXXX sources: `OriginalData.csv`, `factor_loadings.csv`, `entropy_weights.csv`,
`model_comparison_metrics.csv`, `DATA_SOURCE.txt`

Data. $N = 50$, $p = 15$ Likert rows in `OriginalData.csv`. The file is `SYNTHETIC_FALLBACK` (seed 2020). GitHub HiMCM2020 is 400 rows $\times$ 7 unnamed scores, not this 15-item table.

Suitability. KMO = **0.445** (below 0.70). Bartlett $\chi^2 = \mathbf{170.05}$, $\text{df} = 105$, $p = \mathbf{6.10 \times 10^{-5}}$. We reject $R = I_{15}$. We do *not* report KMO 0.782.

Latent factors. PC extraction + Kaiser-normalized Varimax, contest $m = 3$. Eigenvalues 3.184, 1.645, 1.566. Share of trace **42.6%** (not 71.4%). Kaiser $\lambda > 1$ keeps $m = 6$. F1 = network-skill latent factor (safety, network, skill, tips). F2 = autonomy-access latent factor. F3 = burden latent factor (physical +; boss fairness –). Contest names financial yield / human capital / ergonomic burden do not match Λ^* .

Entropy weights (`entropy_weights.csv`). $\mathbf{w} = (0.479, 0.319, 0.202)$. Utility $U = w_1 F_1 + w_2 F_2 - w_3 F_3$. Tutor ranks first ($U = +0.213$). Lifeguard ranks last ($U = -0.681$). We do not use $w = (0.412, 0.354, 0.234)$.

Clusters (`cluster_centroids.csv`). $K = 3$ by contest rule. Silhouette at $K = 3$ is 0.260. Elbow prefers $K = 2$. Max silhouette is $K = 7$. Sizes 23/10/17. Cluster 2 is not a low-burden group (template SSE 2.28).

Classifiers (LOOCV, 50 folds). Features: Thompson $F \in \mathbb{R}^{50 \times 3}$ only. Softmax Top-1 **0.20**, Top-3 **0.38**, log-loss 2.519. ReLU MLP Top-1 0.20, Top-3 **0.48**, log-loss 4.865. Majority Top-1 **0.22**. We do *not* report Top-3 92% vs 84%.

Tool. Streamlit maps 1–10 sliders onto student min–max of F , then Softmax. Pragmatist preset (9,5,8): Retail Clerk 41.8%. Treat ranks as preference, not a job offer.